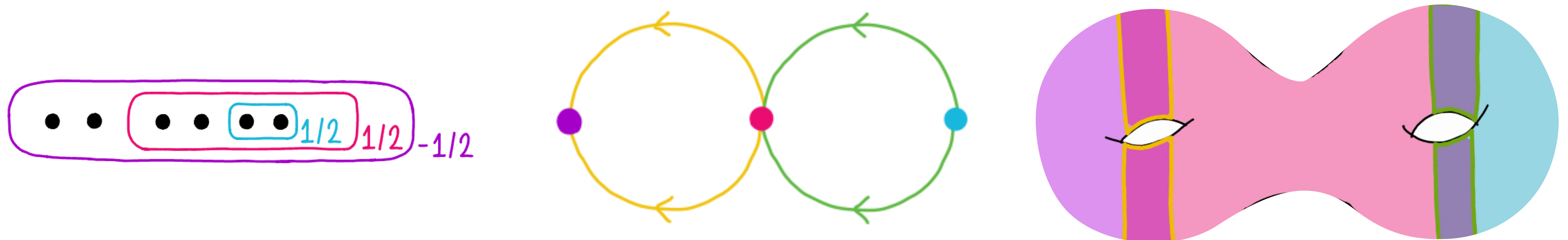


Local heights away from p for quadratic Chabauty

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Boston University $\longrightarrow$ University of Groningen

Joint work with Alexander Betts, Sachi Hashimoto, and Pim Spelier.



Quadratic Chabauty

Theorem [Balakrishnan—Dogra, '18]. Let C be a nice curve of genus g and J be its Jacobian with Mordell-Weil rank r . Suppose that $r = g (+\epsilon)$, and that $Z \subset C \times C$ is a trace 0 correspondence fixed by the Rosati involution. Then there exists a quadratic function

$$\eta_Z: \operatorname{Lie} \left(J_{\mathbb{Q}_p} \right) \rightarrow \mathbb{Q}_p$$

for which $C(\mathbb{Q})$ is contained in the locus inside $C(\mathbb{Q}_p)$ cut out by the equation

$$\eta_Z(\log(z)) - h_{Z,p}(z) \in \Omega,$$

$$\text{where } \Omega = \left\{ \sum_{\ell \neq p} h_{Z,\ell}(x_\ell) : x_\ell \in C(\mathbb{Q}_\ell) \right\}.$$

Quadratic Chabauty and heights

Key input: Let p be a prime number and $Z \subset C \times C$ be a trace 0 correspondence fixed by the Rosati involution. There the associated p -**adic** (Coleman—Gross) **height function** $h_Z : C(\mathbb{Q}) \rightarrow \mathbb{Q}_p$ can be decomposed as

$$h_Z(Q) = \sum_{\ell} h_{Z,\ell}(Q),$$

where $h_{Z,\ell} : C(\mathbb{Q}_{\ell}) \rightarrow \mathbb{Q}_p$.

- For $\ell \neq p$ the height function $h_{Z,\ell}(x_{\ell})$ takes only finitely many values for $x_{\ell} \in C(\mathbb{Q}_{\ell})$.
- If $\ell \neq p$ is a prime of potential good reduction, then $h_{Z,\ell}(x_{\ell}) = 0$ for all $x_{\ell} \in C(\mathbb{Q}_{\ell})$.

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Today's problem: to compute heights $h_{Z,\ell}$ for $\ell \neq p$ odd of bad reduction.

Quadratic Chabauty and heights

Many of the applications of quadratic Chabauty so far concern curves

- with trivial local height contributions away from p ; or
- whose special fibre at primes of bad reduction consists of a unique irreducible component; or
- for which one can use elliptic curve factors.

Today's problem: to compute heights $h_{Z,\ell}$ for $\ell \neq p$ odd of bad reduction.

Theorem [Betts—D.—Hashimoto—Spelier, '25]. Let $C/\mathbb{Q}$ be a hyperelliptic curve that admits a model $y^2 = f(x)$, where $f(x)$ is separable and of degree ≥ 3 . Let $Z \subset C \times C$ be a trace 0 correspondence fixed by the Rosati involution. Let p be a prime number. Then there is a provably correct algorithm to compute the function $h_{Z,\ell}$ for all odd primes of bad reduction $\ell \neq p$.

Explicit height computations: how?

Theorem [Betts—Dogra, '20]. Let p be a prime number and $Z \subset C \times C$ be a trace 0 correspondence fixed by the Rosati involution. Then, there is an explicit formula for computing $h_{Z,\ell}$ in terms of:

- the induced action of Z_* on the homology $H_1(\Gamma, \mathbb{Q})$ of the dual graph Γ of the geometric special fibre, and
- the traces $\text{tr}_v(Z)$ attached to vertices of Γ .

This formula uses a semistable model of C .

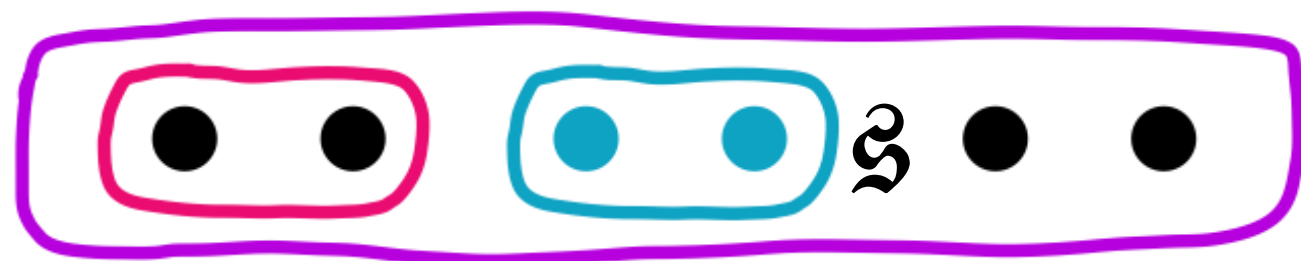
Explicit height computations: how?

We need to understand:

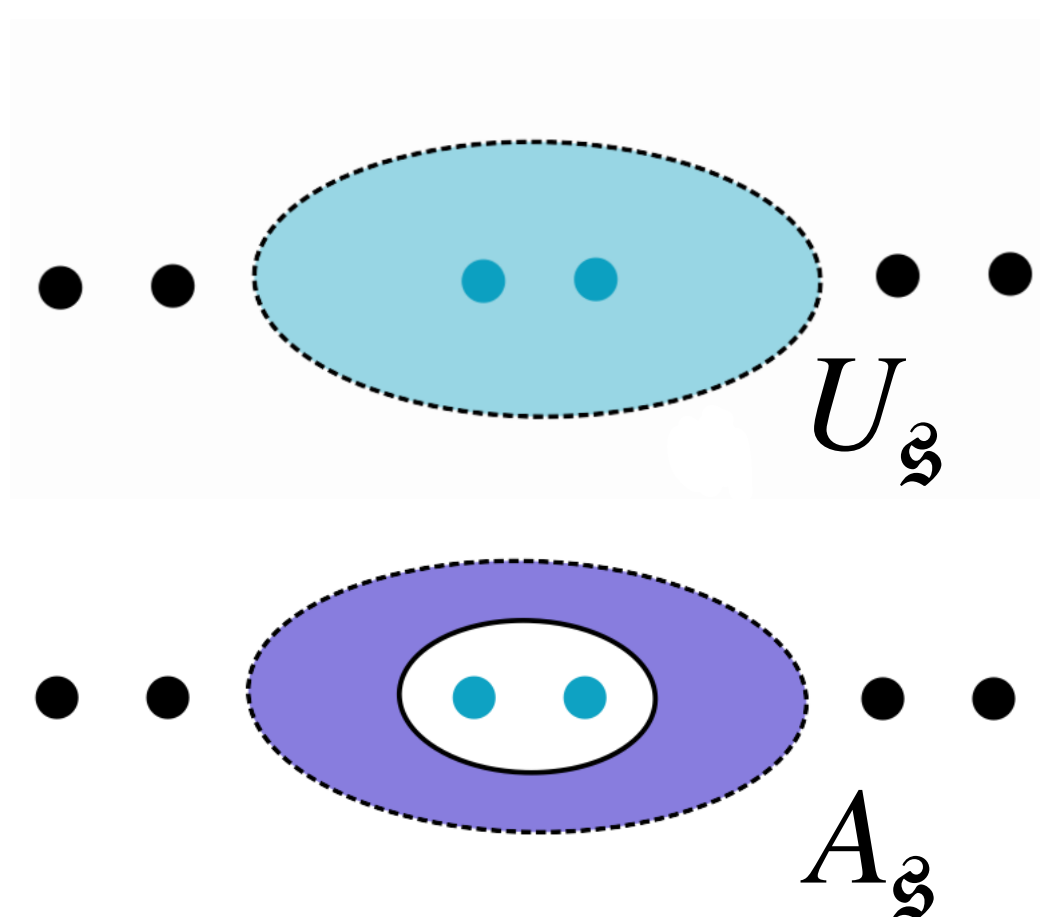
1. A semistable model of C .
2. The induced action of Z_* on the homology $H_1(\Gamma, \mathbb{Q})$ of the dual graph Γ of the geometric special fibre.
3. The traces $\mathrm{tr}_v(Z)$ attached to vertices of Γ .

1. A semistable covering

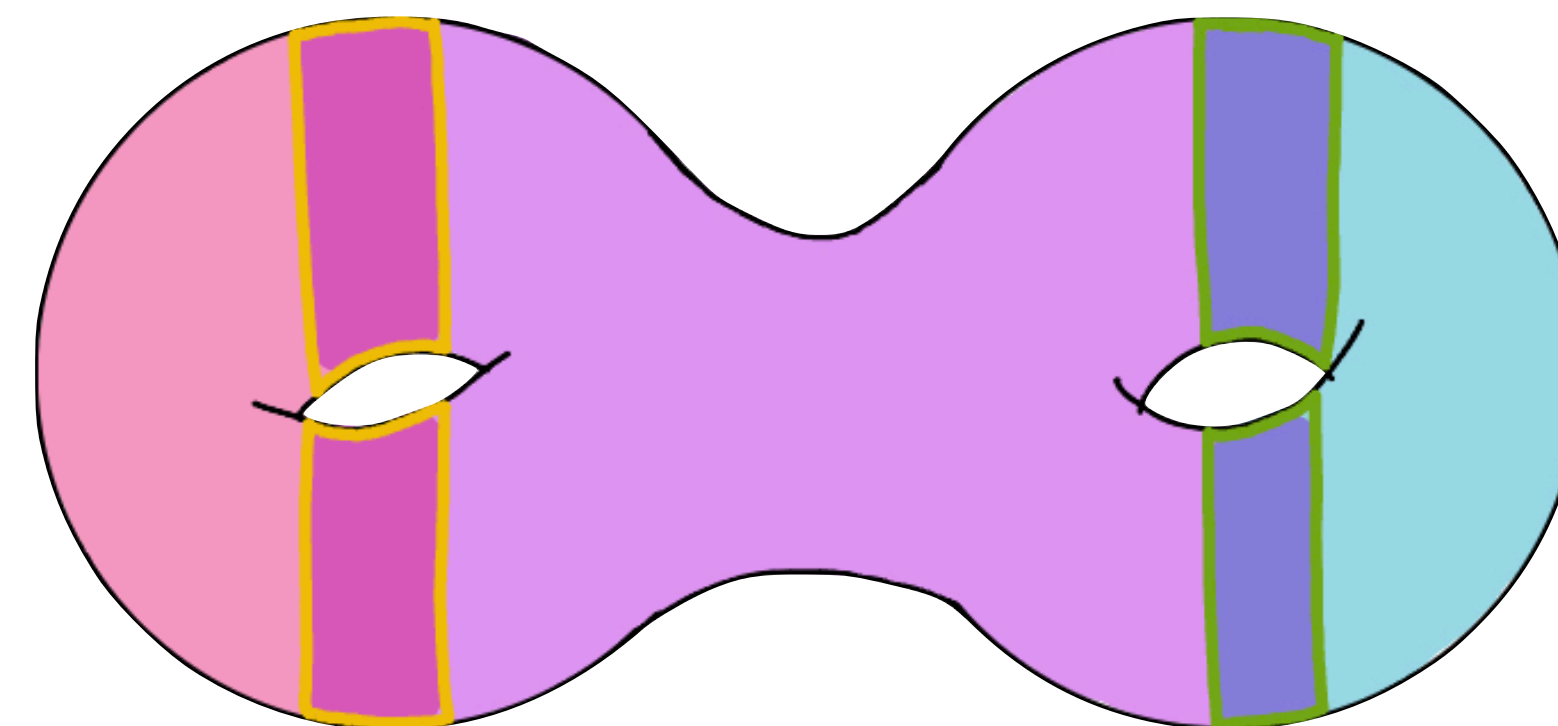
Instead of finding a semistable model of C (or work explicitly with them), we construct a semistable covering of C using cluster pictures [DDMM '23].



Cluster picture:
roots of $f(x)$.



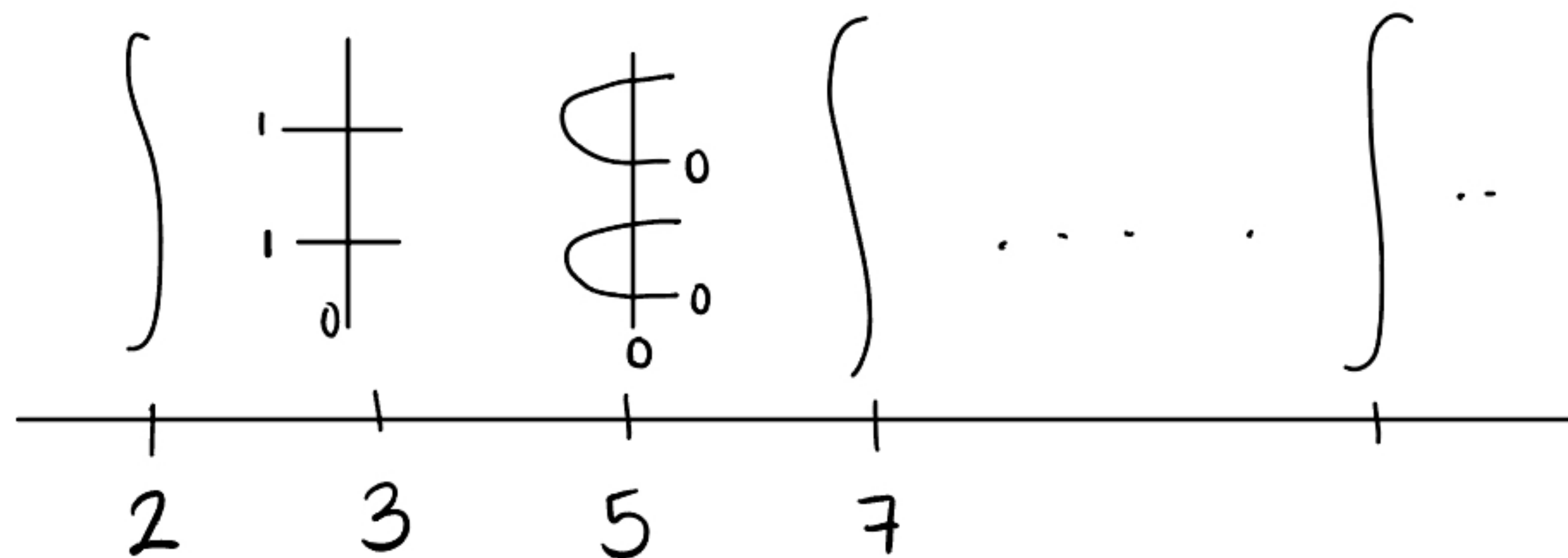
Semistable
covering of $\mathbb{P}^{1,\text{an}}$



Semistable
covering of C^{an}

Example

$$C : y^2 = x^6 + \frac{18}{5}x^4 + \frac{6}{5}x^3 + \frac{9}{5}x^2 + \frac{6}{5}x + \frac{1}{5}$$



Example

$$C : y^2 = x^6 + \frac{18}{5}x^4 + \frac{6}{5}x^3 + \frac{9}{5}x^2 + \frac{6}{5}x + \frac{1}{5}, \quad p = 5$$

```
> R<x> := PolynomialRing(pAdicField(5));  
> f:= x^6 + 18/5*x^4 + 6/5*x^3 + 9/5*x^2 + 6/5*x + 1/5;  
> Factorization(f);  
[  
  <(5 + 0(5^21))*x + 2068953077662*5 + 0(5^20), 1>,  
  <(5^2 + 0(5^22))*x^2 + (6373813938821*5^2 + 0(5^21))*x + 343719541919*5^2 +  
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  <(5 + 0(5^21))*x - 1185825657571*5 + 0(5^20), 1>,  
  <(5^2 + 0(5^22))*x^2 - (7256941358912*5^2 + 0(5^21))*x - 3436100256277*5 +  
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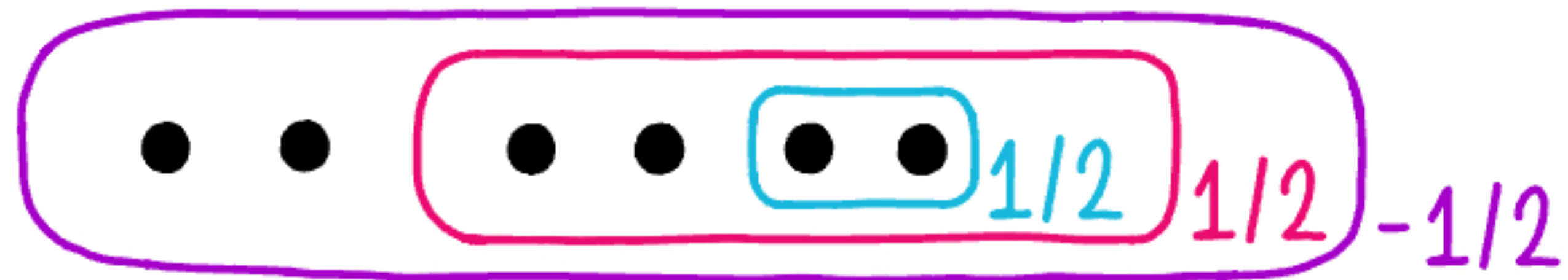
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```
> fL := SemistableField(f, p : prec := 30, printing := false);  
> roots := Roots(fL);  
> [Valuation(roots[1][1]-roots[i][1]) : i in [2..6]];  
[ 0, 0, 0, -2, -2 ]  
> [Valuation(roots[2][1]-roots[i][1]) : i in [1..6]|i ne 2];  
[ 0, 2, 0, -2, -2 ]
```

Example

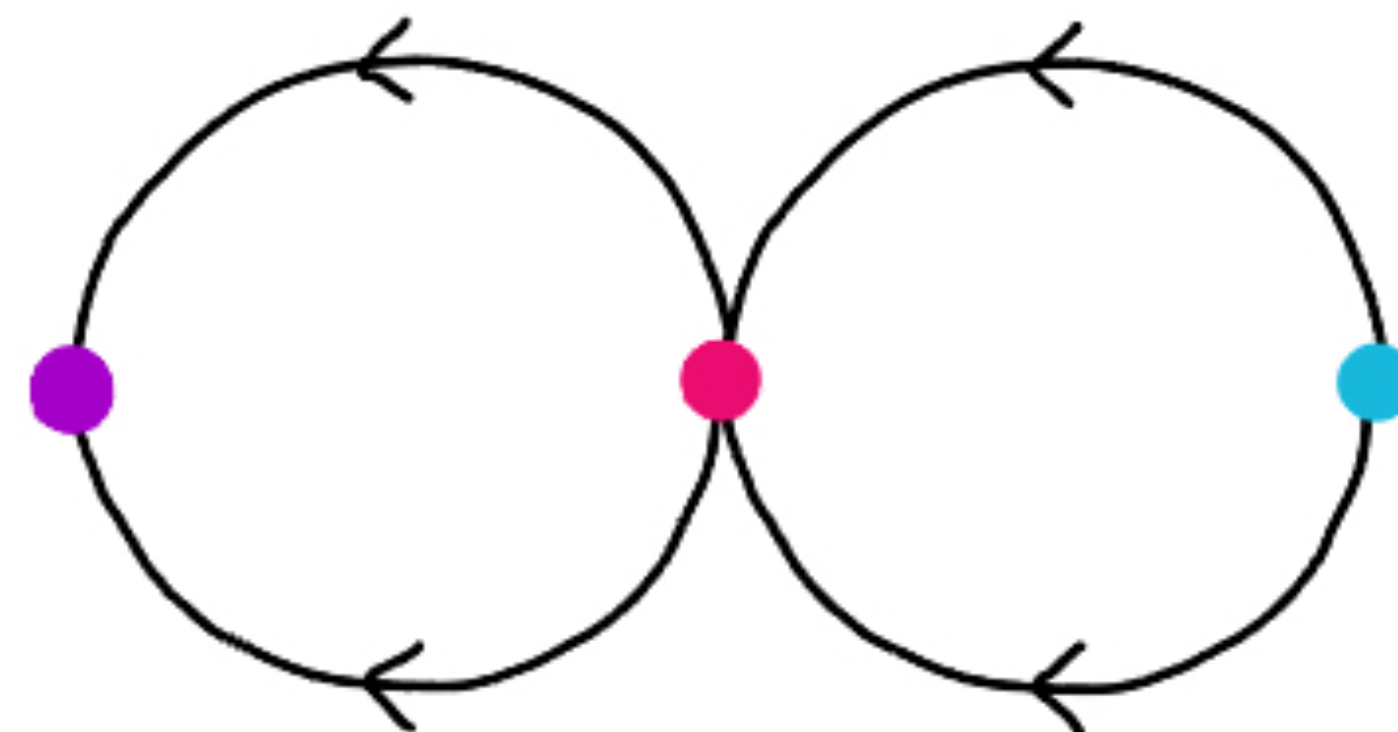
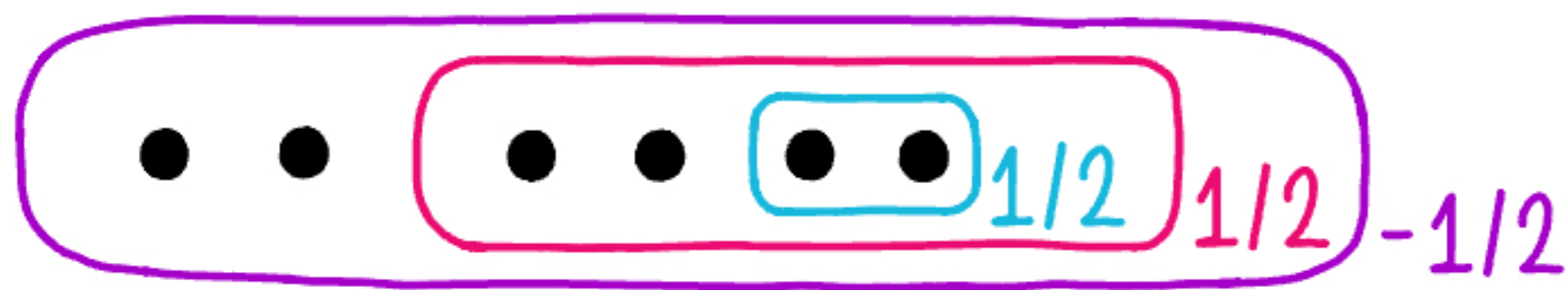
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Example

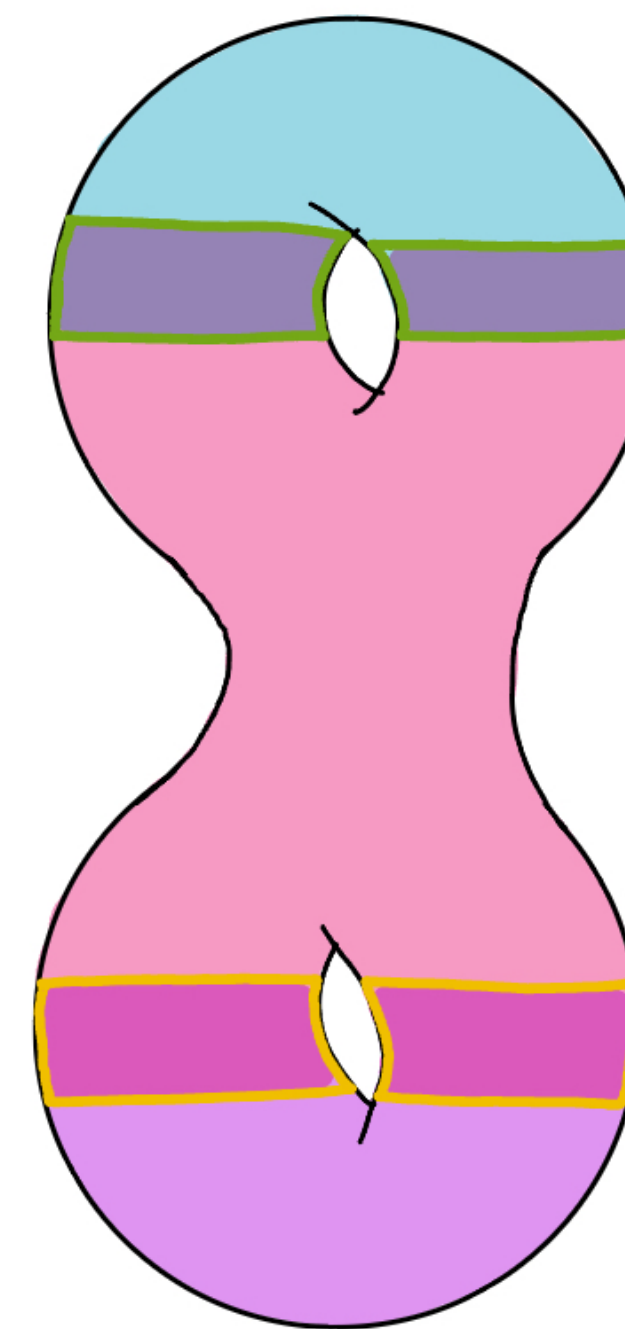
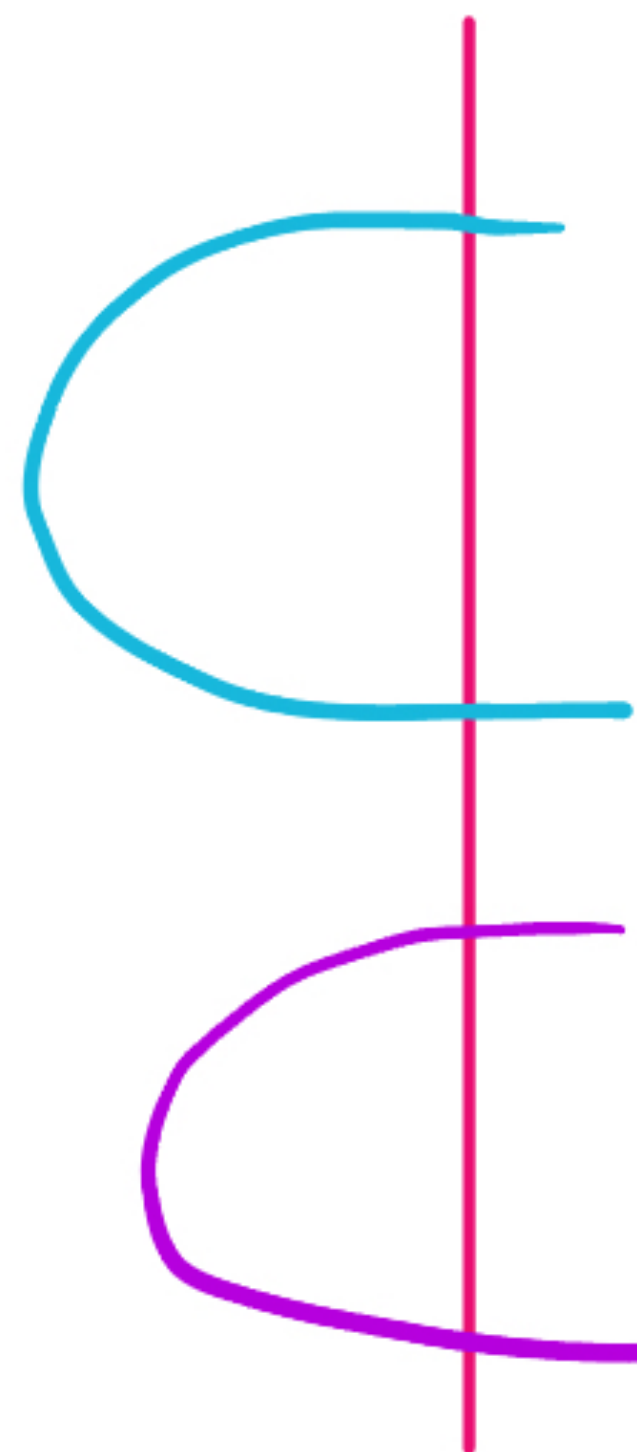
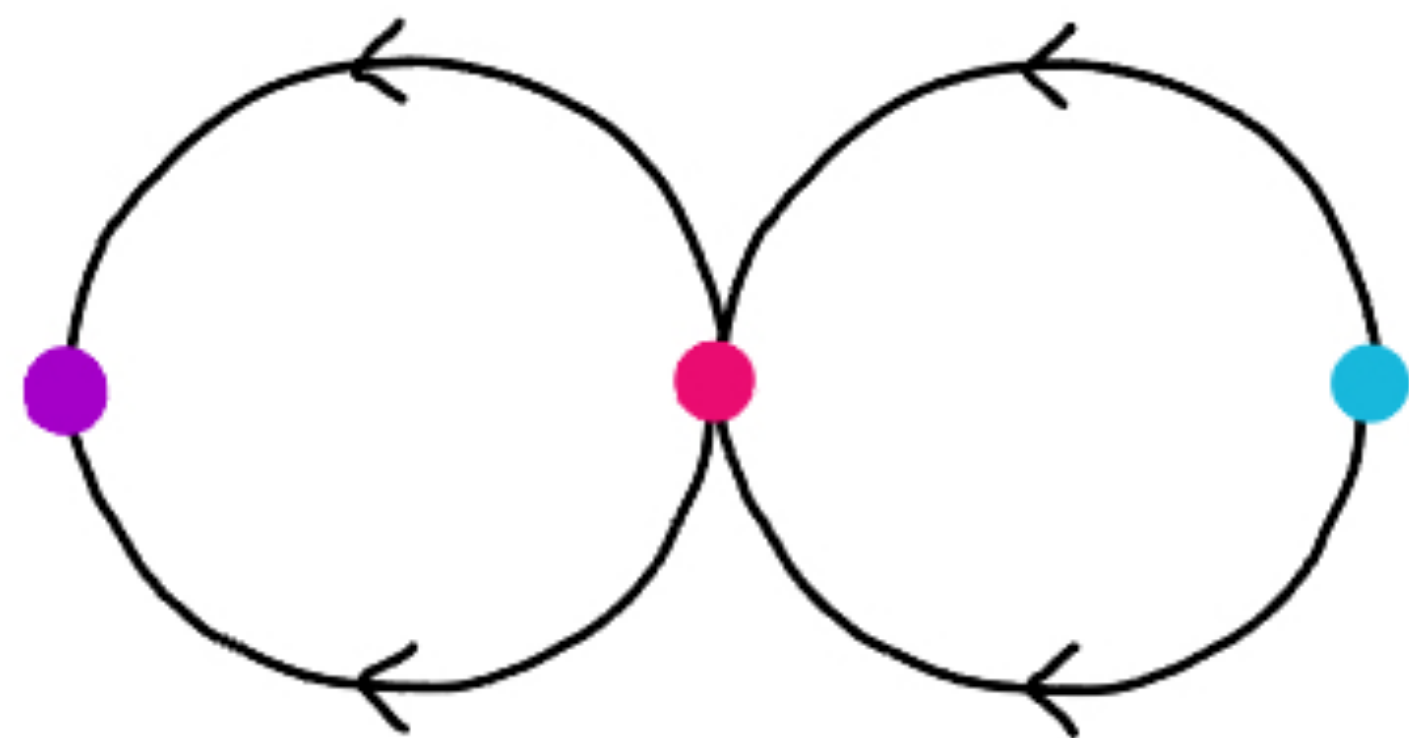
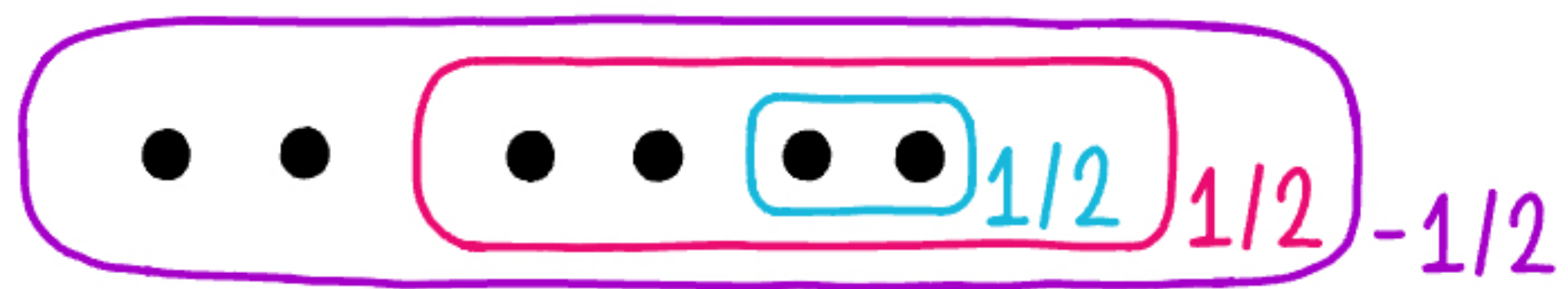
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$$p = 5$$



2. The action of Z_* on $H_1(\Gamma, \mathbb{Q})$: Coleman - Iovita

We represent Z as a divisor in $C \times C$ [Costa—Mascot—Sijssling—Voight, '19].

This allows us to understand the action of Z_* on $H^0(C_{\mathbb{Q}_\ell}, \Omega_C^1)$.

Theorem [Coleman & Iovita, '99]. The map

$$H^0(C_{\mathbb{Q}_\ell}, \Omega_C^1) \rightarrow H_1(\Gamma, \mathbb{Q}_\ell), \text{ given by } \omega \mapsto \sum_{e \in E(\Gamma)} \text{Res}_{A_{\vec{e}}}(\omega) \cdot \vec{e},$$

is surjective. Moreover, the map is an isomorphism if every component of the ℓ -adic special fibre has genus 0.

We can compute the action of Z_* on $H_1(\Gamma, \mathbb{Q}_\ell)$ up to any desired ℓ -adic precision.

Coleman - lovita done explicitly

Theorem [Betts—D—Hashimoto—Spelier, '25]. The endomorphism of $H_1(\Gamma, \mathbb{Q})$ given by Z_* is defined over $\mathbb{Z}$, and has operator norm $\leq \sqrt{d_1 d_2}$, where d_1 and d_2 are the degrees of the two projections $Z \rightarrow C$.

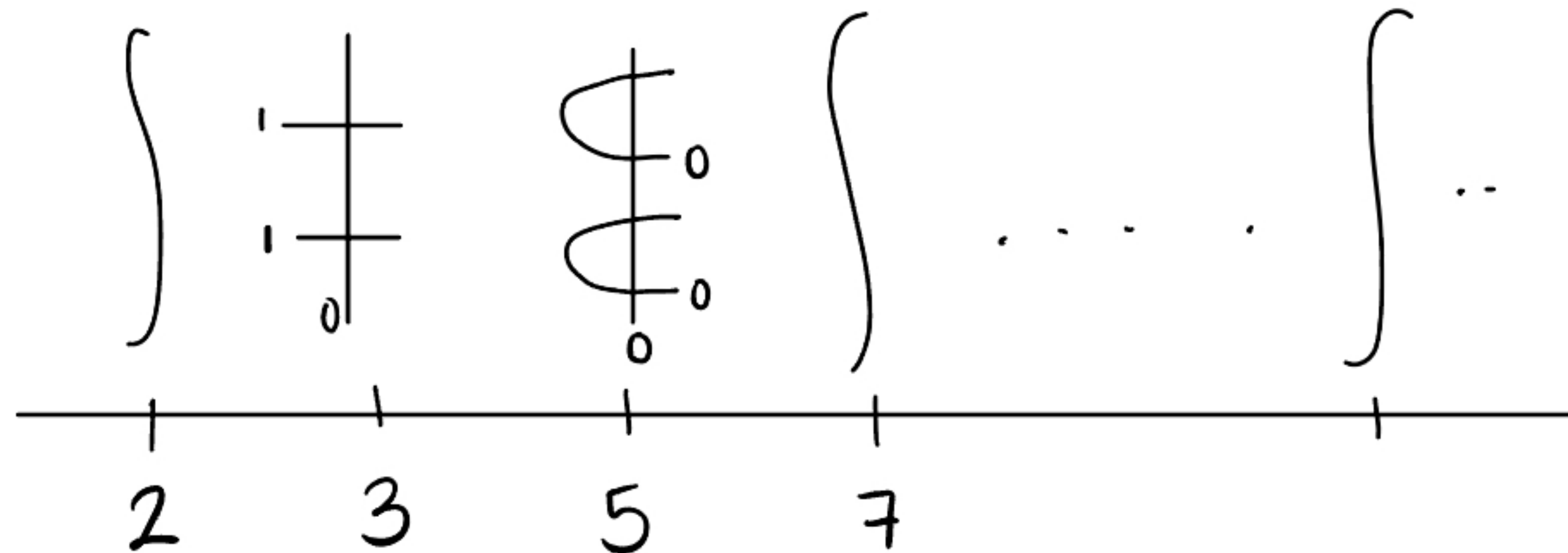
Finite precision is enough!

Local heights computations: an example

$$C : y^2 = x^6 + \frac{18}{5}x^4 + \frac{6}{5}x^3 + \frac{9}{5}x^2 + \frac{6}{5}x + \frac{1}{5}$$

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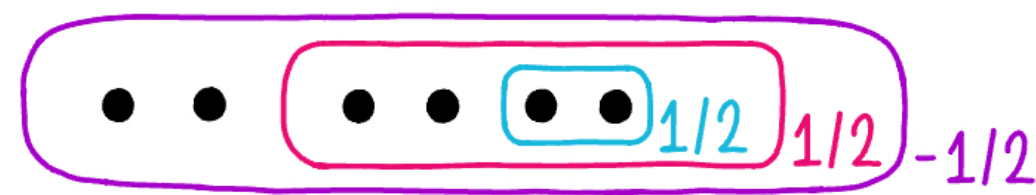
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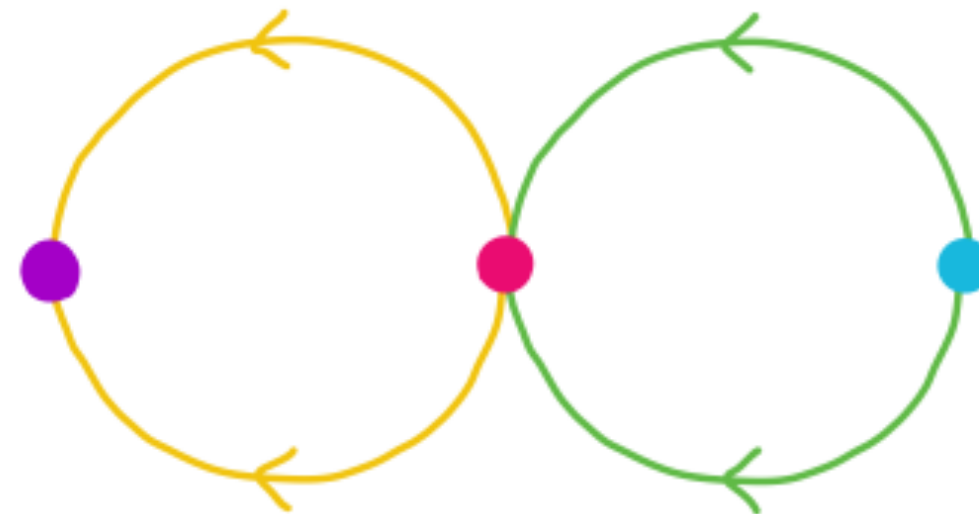
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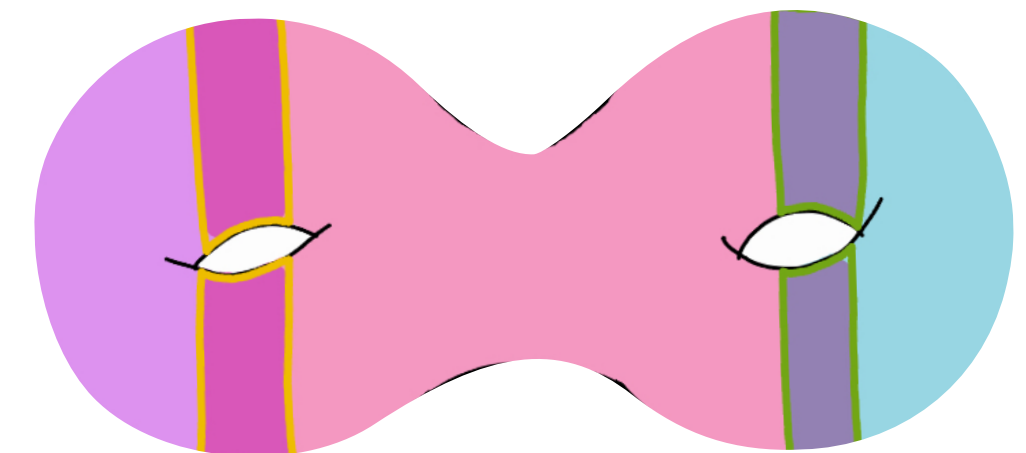
Pick the correspondence Z from the endomorphism $\sqrt{13}$ on the Jacobian. Let $\ell = 5$.



Cluster picture



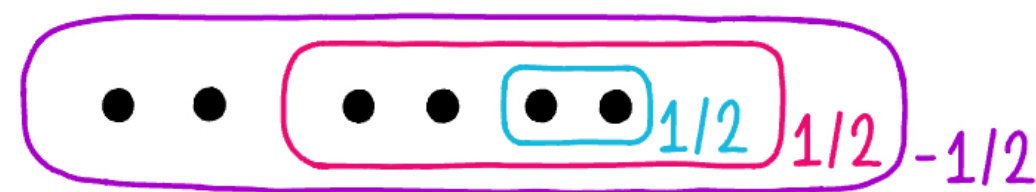
Berkovich space decomposition



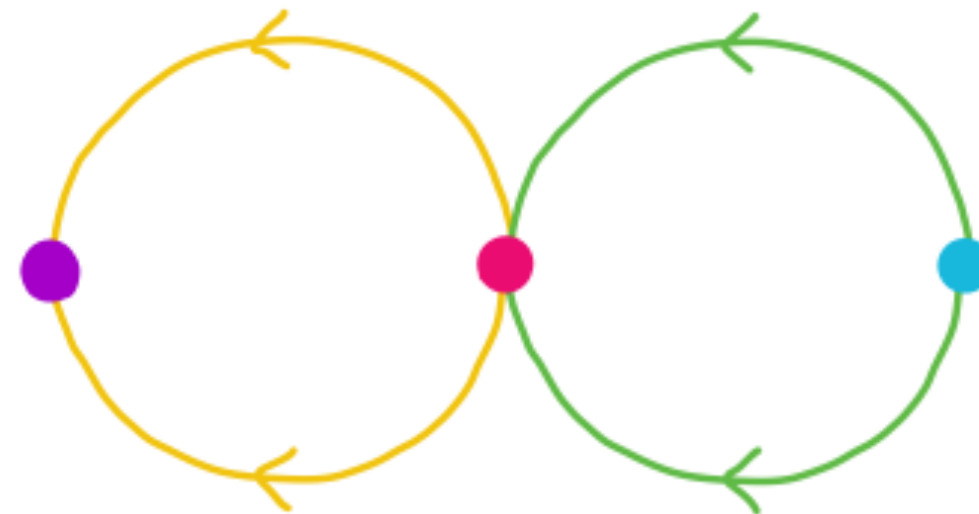
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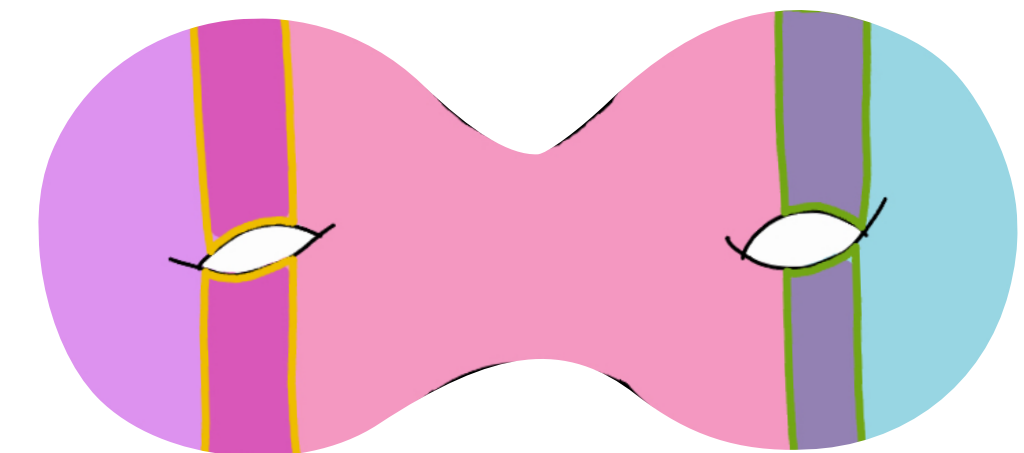
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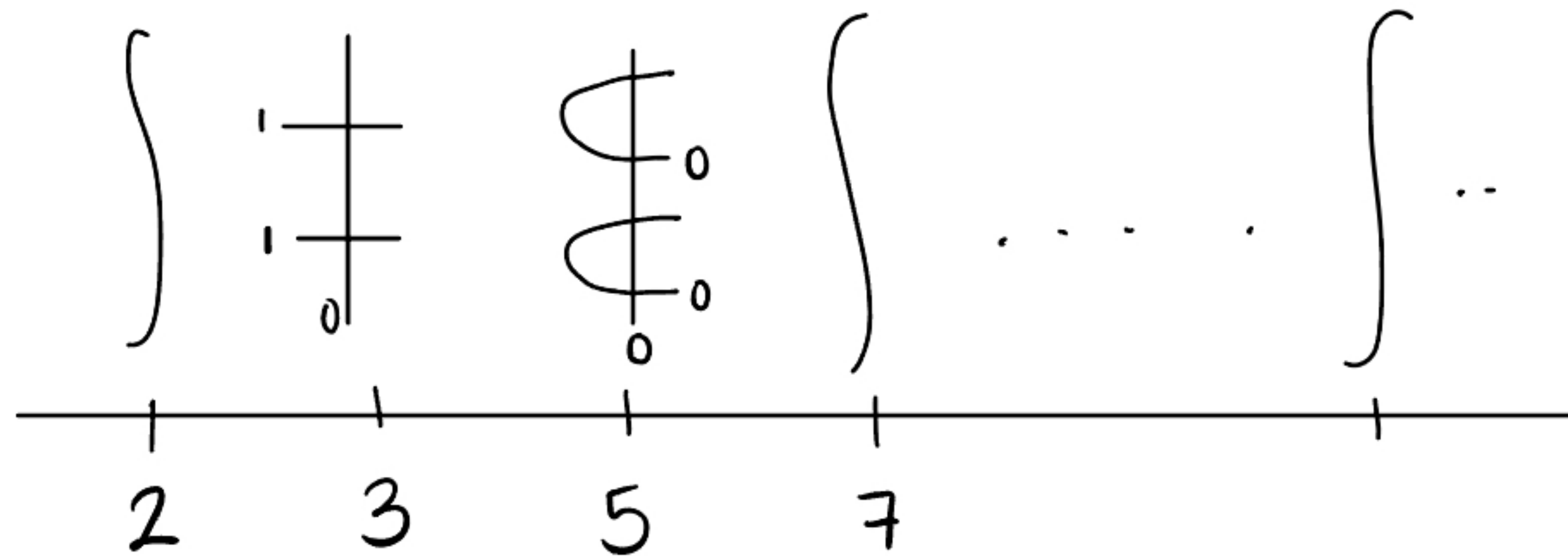
Berkovich space decomposition



$$h_{Z,5}(x : y : z) = \begin{cases} +\frac{3}{4} \log_p(5) & \text{if } z \equiv 0 \pmod{5}, \\ -\frac{3}{4} \log_p(5) & \text{if } x/z \equiv 2 \pmod{5}, \\ 0 & \text{otherwise.} \end{cases}$$

Local heights computations: an example

$$C : y^2 = x^6 + \frac{18}{5}x^4 + \frac{6}{5}x^3 + \frac{9}{5}x^2 + \frac{6}{5}x + \frac{1}{5}$$



$$h_{Z,3}(x : y : z) = \begin{cases} 0 & \text{if } z \equiv 0 \pmod{3}, \\ -\frac{2}{3} \log_p(3) & \text{if } x/z \equiv 1 \pmod{3}. \end{cases} \quad h_{Z,5}(x : y : z) = \begin{cases} +\frac{3}{4} \log_p(5) & \text{if } z \equiv 0 \pmod{5}, \\ -\frac{3}{4} \log_p(5) & \text{if } x/z \equiv 2 \pmod{5}, \\ 0 & \text{otherwise.} \end{cases}$$

Application: quadratic Chabauty

$$C : y^2 = x^6 + \frac{18}{5}x^4 + \frac{6}{5}x^3 + \frac{9}{5}x^2 + \frac{6}{5}x + \frac{1}{5}$$

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Theorem [Betts—D—Hashimoto—Spelier, '25]. The rational points on C are the 10 points:

$$\{(-1/3 : -1/27 : 1), (-1/3 : 1/27 : 1), (-1/5 : -21/125 : 1), (-1/5 : 21/125 : 1), (1 : 3 : 1), (1 : -3 : 1), (1 : 1 : 0), (1 : -1 : 0), (-1/2 : -3/8 : 1), (-1/2 : 3/8 : 1)\}$$

Our method

Our implementation handles $\ell \neq p, 2$ for any hyperelliptic curve that:

1. is given by an affine model $y^2 = f(x)$, where $f(x)$ is a separable polynomial and has even degree > 3 ;
2. its Jacobian has a nontrivial endomorphism.

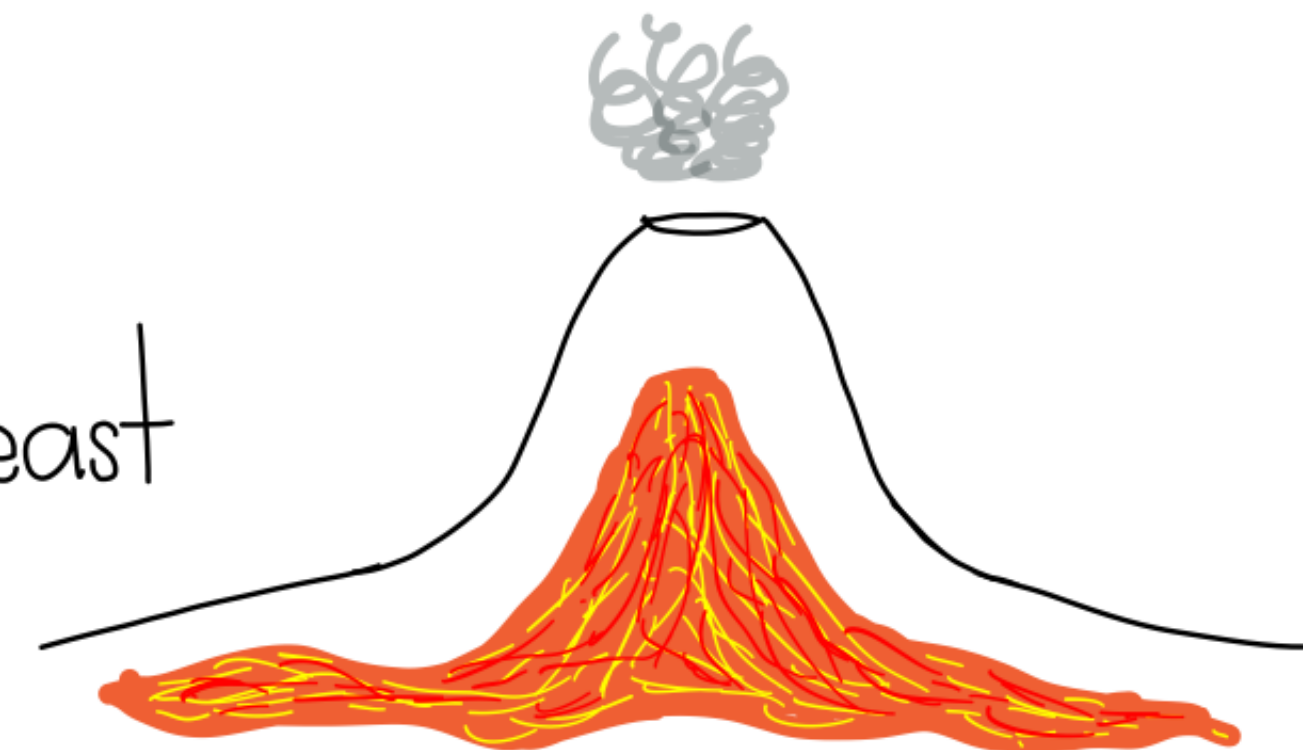
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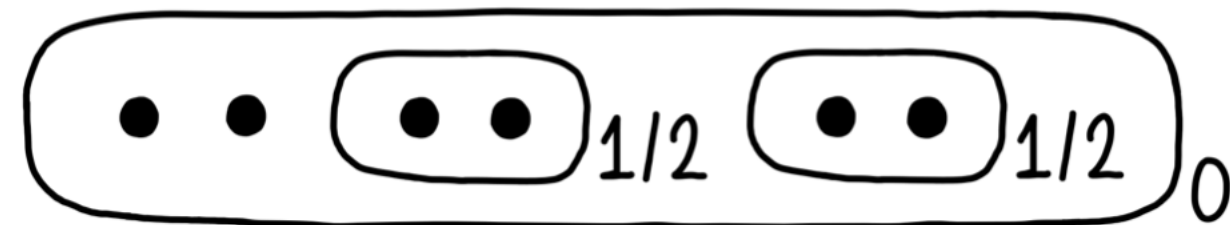


available at your neareast



Consequences of our method

- If $C/\mathbb{Q}_\ell$ is a genus 2 curve with the cluster picture



and with leading coefficient a unit in $\mathbb{Z}_\ell$, then the local height of any $\mathbb{Q}_\ell$ -point on C is 0.

- The Shimura curve quotient $X_0(93,1)/\langle \omega_{93} \rangle$ has trivial local height at 31.
- The Atkin-Lehner quotients $X_0(330)^*$, $X_0(255)^*$, and $X_0(147)^*$ have trivial local heights away from p .

Generalization: plane quartic

$$xy^3 + 2xy^2 + (x - x^3 - p)y - p = 0$$

Recall that we need to understand:

- | | |
|---|---|
| 1. A semistable covering of C . | Cluster pictures. |
| 2. The induced action of Z_* on the homology $H_1(\Gamma, \mathbb{Q})$ of the dual graph Γ of the geometric special fibre. | Coleman-Iovita. |
| 3. The traces $\mathrm{tr}_v(Z)$ attached to vertices of Γ . | Local/global symbols and push/pull formula. |

Generalization: plane quartic

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Recall that we need to understand:

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3 to 1 cover of $\mathbb{P}^1$.
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We follow ideas of Helminck '21 and Ossen '24.

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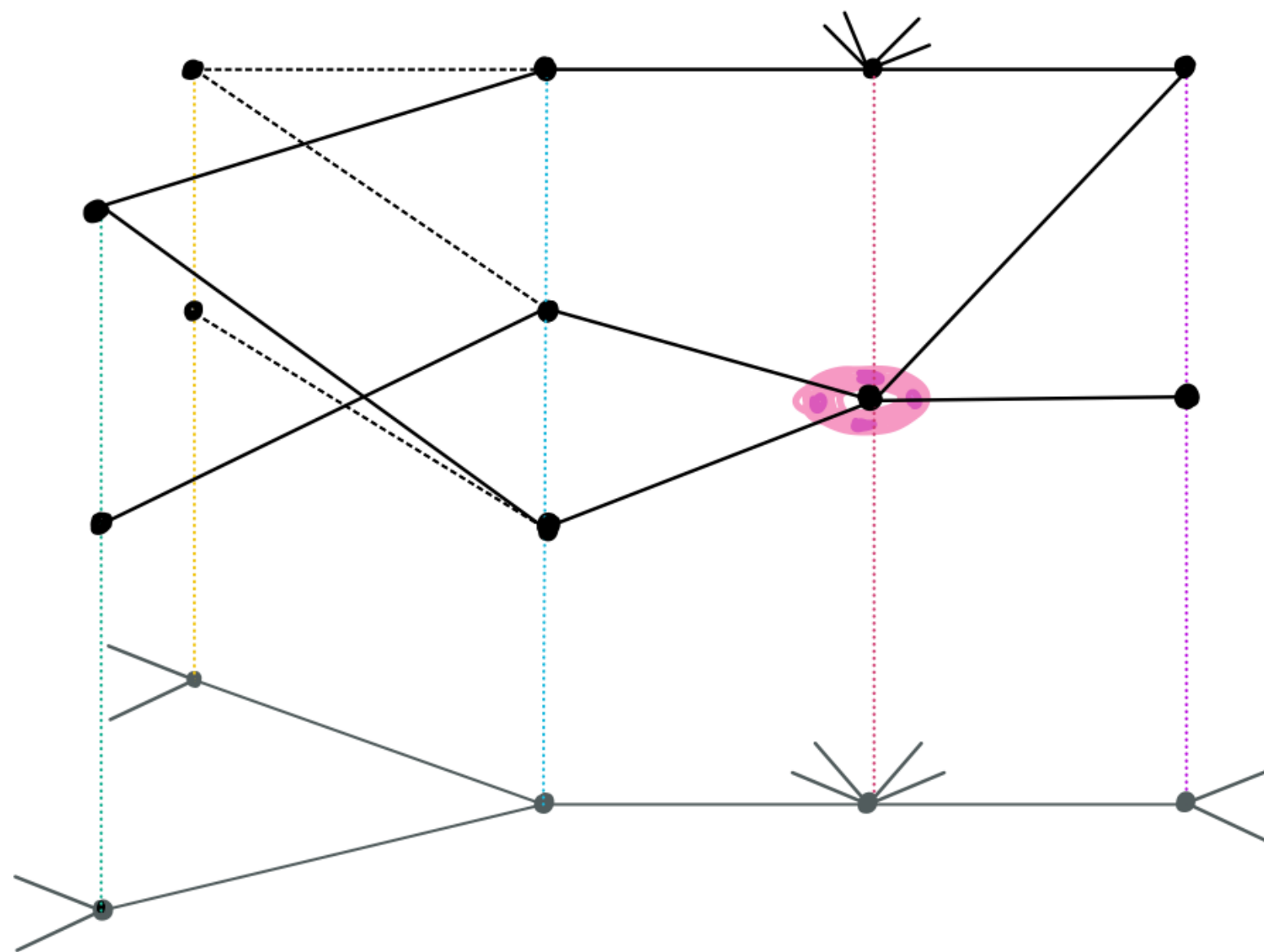
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We follow ideas of Helminck '21 and Ossen '24.



Thank you!

To compute $h_{Z,\ell}$ for hyperelliptic curves, we find

1. A semistable covering of C .
Cluster pictures.
2. The induced action of Z_* on the homology $H_1(\Gamma, \mathbb{Q})$ of the dual graph Γ of the geometric special fibre.
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